

## *Issac Tabares*

### My Physics of Time and Infinity

An Interpretive Research Essay on Time, Entropy, Relativity, Quantum Theory, Infinity, Everettian Unified Duality, Stable Records, Relative Time, and Consciousness-First Ontology

#### Thesis

I argue for **Everettian unified duality** with a **consciousness-first ontology**, and I separate what physics supports from what I personally commit to. The physics spine is time, entropy, relativity, quantum theory, cosmology, information bounds, decoherence, stable records, and the limits of infinities. The metaphysical claim is that finite systems are locally real but not absolutely separate from the totality that contains them.

By **consciousness-first**, I mean the metaphysical thesis that first-person interiority is ontologically fundamental rather than derived from an entirely non-experiential reality. First-person interiority is the subjective presence of experience; **proto-experiential interiority** names a proposed minimal interior aspect without thereby attributing perception, thought, memory, or a self to it. Physics describes symmetries, states, dynamics, correlations, and records; how that organization relates to experience remains unresolved. In my framework, consciousness is neither a new physical force nor something that collapses the wave function, and quantum mechanics does not establish that it is fundamental.

I was shaped by many scientists and writers, but **Sean Carroll** has been the constant learning source for me for most of my life. His work on time, entropy, relativity, quantum foundations, and Hilbert-space realism gave me the language to discipline questions I already had about existence.

By **unified duality**, I mean local distinction together with global belonging. **Finite configurations** have real differences, causal histories, and boundaries of their own, even when those boundaries are effective and approximate. Belonging to a larger structure does not erase those differences. **Everettian unified duality** applies this relationship to an interpretation of quantum mechanics that accepts decohered branches as physically real structures within the universal state. That state describes physical totality; it is not equivalent to **Ultimate Totality**, which also includes nonphysical modes in my ontology. I reserve **Absolute Consciousness** for the interior aspect I metaphysically attribute to Ultimate Totality. The term does not describe an enlarged human observer or turn global belonging into proof of a universal subject.

## Claim Status

| Claim                                   | Status                                             | Support                                                                                                                                                                           | Limit                                                                                    |
|-----------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Time is layered                         | Established physics plus open quantum-gravity work | Proper time, thermodynamic records, quantum time as a parameter, relational-clock models, and semiclassical time recovery.                                                        | No final theory of quantum gravity is established.                                       |
| Finite local configurations             | Claim compatible with physics                      | Decoherence, density matrices, effective field theory, and entropy bounds describe bounded local systems inside wider structures.                                                 | This does not prove the final ontology of the whole.                                     |
| Locality may be emergent                | Serious unresolved research                        | Quantum-information approaches use entanglement and mutual information to reconstruct geometry in controlled settings.                                                            | No complete derivation of our universe from Hilbert space is established.                |
| Everettian branches are real            | Interpretive physics stance                        | Unitary quantum mechanics plus decoherence explains stable branch-like histories without adding physical collapse.                                                                | Everettian ontology and probability remain interpretive issues.                          |
| Observer boundaries require records     | Physics-facing interpretive claim                  | A finite observer must be a stable, localized, record-bearing subsystem, not an arbitrary mathematical slice.                                                                     | The exact bridge between physical records and subjective interiority remains incomplete. |
| First-person interiority is fundamental | Metaphysical commitment                            | Physics correlates experience with physical structure but does not currently derive first-person presence from non-experiential matter.                                           | No accepted experiment proves this ontology.                                             |
| Ultimate totality                       | Personal metaphysical commitment                   | Ultimate Totality contains all modes of form: physical, possible, mathematical, conceptual, counterfactual, and counterpossible. Only the physical subset must obey physical law. | No observer can step outside reality and measure the whole as an external object.        |

## Physics and Math Spine

The equations in this work make the claims and their assumptions more precise. Mathematical consistency alone does not establish a physical hypothesis: its domain of validity and comparison with observations also matter. Research models can yield conditional results without being confirmed theories. I distinguish those results from interpretations of the formalism and from my personal **ontology**.

The quantum starting point is a state in Hilbert space evolving under a Hamiltonian. For a closed system with time-independent Hamiltonian, the clean unitary form is

$$U(t) = e^{-iHt/\hbar}.$$

Unitary evolution preserves the norm of the global state without introducing physical collapse. Decoherence also requires interactions that correlate subsystems with their environment. Branch structure is identified effectively through stable histories and records; attributing ontological reality to those branches is the Everettian interpretive step.

The finite-information side starts from black-hole thermodynamics. The central result is that black-hole entropy scales with horizon area:

$$S_{BH} = \frac{k_B c^3 A}{4G\hbar}.$$

Here  $S_{BH}$  is the entropy of the black hole,  $A$  is the horizon area,  $G$  is Newton's gravitational constant,  $c$  is the speed of light,  $\hbar$  is the reduced Planck constant, and  $k_B$  is Boltzmann's constant. The important point is that entropy scales with area, not volume.

The following bound requires interpreting maximum entropy as a bound over all admissible states in the region's accessible state space, with its physical restrictions specified. Finite entropy in a particular state does not suffice to bound the Hilbert-space dimension. Under that entropy-counting assumption:

$$\dim \mathcal{H}_R \lesssim e^{S_{\max}/k_B}.$$

Gravitational arguments about entropy bounds motivate a **finite information capacity** for regions under appropriate physical conditions. The proposal of a finite local dimension requires those additional assumptions; it is not a general consequence of finite spatial extent. It offers a reason to investigate whether the continuum of quantum field theory is an approximation rather than the final **ontology**.

Quantum information provides a way to describe correlations between subsystems. One measure of those correlations is **quantum mutual information**:

$$I(A : B) = S(A) + S(B) - S(AB).$$

Here  $S(A)$ ,  $S(B)$ , and  $S(AB)$  are von Neumann entropies of subsystem  $A$ , subsystem  $B$ , and their joint system. In specific models, entanglement and information relations help reconstruct aspects of geometry and locality. Mutual information is a tool for these reconstructions, but does not by itself define a general spatial distance. Whether our geometry can be recovered from a deeper quantum structure without presupposing all of its spatial organization remains open.

A subsystem is described by taking the partial trace over environmental degrees of freedom. This operation defines its reduced state; it is not, by itself, a decoherence process:

$$\rho_S = \text{Tr}_E(\rho_{SE}).$$

Decoherence occurs when interaction with the environment effectively suppresses interference between alternatives relevant to local observables. Under suitable conditions, it supports robust records and an approximately classical structure. This explains aspects of branch stability without selecting a single global outcome or, by itself, establishing the ontological reality of branches.

## Types of Time

| Type               | Where It Appears                                             | Status                          | Meaning                                                    |
|--------------------|--------------------------------------------------------------|---------------------------------|------------------------------------------------------------|
| Coordinate time    | Special relativity and general relativity coordinate systems | Conventional                    | A label used inside a chosen chart.                        |
| Proper time        | Special relativity and general relativity worldlines         | Invariant and local             | What an idealized local clock measures along its own path. |
| Cosmic time        | Friedmann-Lemaitre-Robertson-Walker cosmology                | Symmetry dependent              | Proper time of comoving observers.                         |
| Thermodynamic time | Statistical mechanics                                        | Emergent                        | Direction associated with entropy and records.             |
| Quantum time       | Standard quantum mechanics                                   | External parameter              | Parameter in Schrodinger evolution.                        |
| Relational time    | Page-Wootters and quantum gravity                            | Research and interpretation     | Time reconstructed from correlations.                      |
| Semiclassical time | Wheeler-DeWitt and Wentzel-Kramers-Brillouin limits          | Approximate and emergent        | Time recovered from an almost classical geometry.          |
| Experienced time   | Conscious observers                                          | Philosophical and neurophysical | Lived sequence, memory, rhythm, and expectation.           |

## Boundaries and Influences

This work does not provide a completed theory of consciousness, does not prove that the universe is conscious, does not claim that quantum mechanics proves consciousness, does not claim that atoms think like humans, and does not turn every mathematical infinity into a literal physical infinity.

The careful statement is that physics supports **structural connectedness** more directly than total experiential unity. It gives equations, invariants, correlations, entropy gradients, density matrices, records, and information bounds. The stronger interior claim belongs to the metaphysical side of the work, where I state it as commitment rather than physical proof.

Established or strongly supported physics includes proper time, time dilation, Einstein field equations, entropy increase in macroscopic systems, renormalization practice in quantum field theory, decoherence, and Friedmann-Lemaitre-Robertson-Walker cosmology.

Theoretically serious but unresolved material includes the Past Hypothesis, gravitational entropy, semiclassical time emergence, Page-Wootters relational time, canonical quantum gravity, finite-dimensional Hilbert-space approaches to quantum gravity, emergent locality from entanglement, and quantum circuit cosmology.

The speculative part of my view is consciousness-first ontology, finite configurations inside deeper unity, generative variation, ultimate totality, and the idea that the universe experiences itself through finite perspectives. I believe these ideas are true, but I do not present them as completed experimental physics or as consequences forced by Everett.

## Scientific Influences

The scientific side of this work is shaped by astronomy, equations, light, scale, and the feeling that physical structure points toward questions larger than any single discipline. One early image that stayed with me was an astronomy book from my father's encyclopedia collection: planets, motion, distance, and mathematical order appearing as parts of one connected reality.

The scientific influences that shaped this work most are quantum mechanics, gravitation, statistical mechanics, astrophysics, and quantum chromodynamics. I value them because each one shows reality as structured, layered, and mathematically disciplined instead of obvious or simple.

## Technical Practice and Other Scientific Interests

The website itself is part of my technical practice. Building and correcting it forced me to care about structure, syntax, debugging, readability, accessibility, and the difference between an idea that sounds good and a system that actually works.

Separate from the central argument of this work, I also have active interests in fluid mechanics, geometric optics, black-hole thermodynamics, astrophysics, and human biology through tools such as Complete Anatomy. I do not present those topics as proof for my ontology or as direct support for every claim on this site. I include them because they keep returning me to one broader question: **how can one reality generate structure, time, memory, body, mind, and experience?**

## Personal Cosmology Argument

My framework is not a separate scientific theory added on top of physics. It is the interpretive and metaphysical structure I use after separating established physics, serious open research, and personal commitment. The scientific side supplies constraints: proper time, entropy, decoherence, Hilbert-space structure, density matrices, records, information bounds, cosmological horizons, and the unresolved status of quantum gravity. The metaphysical side is my commitment that first-person interiority is fundamental and that finite perspectives belong to a deeper totality without being erased by it.

The thesis is this: finite observers are locally real as stable, record-bearing configurations, while none of them is absolutely outside the total structure of reality. In Everettian language, a branch-local observer has memories, records, causal history, and embodied limitation inside one decohered history. Those histories remain distinct while belonging to one universal physical state. That is what I mean by **unified duality**: local distinction and global belonging are both real.

The bridge problem is the hard part. Naming consciousness as fundamental does not explain how a human perspective forms. A serious bridge would have to connect intrinsic interiority with physical organization: stable records, memory, attention, embodiment, self-modeling, entropy gradients, and branch-local causal history. The connection cannot be an arbitrary mathematical cut through Hilbert space. A finite observer must be a localized, dynamically stable, causally organized subsystem capable of maintaining robust, time-ordered records of itself and its surroundings.

This is why the ontology remains constrained by physics rather than presented as physics. Structural connectedness, local records, decoherence, gravitational information bounds, and research programs on emergent locality or relational time can motivate the framework. They do not derive fundamental interiority or total experiential unity. Those remain metaphysical commitments.

## Time

Since childhood, I have carried the same questions into every phase of my life: **What is time? Why does it have a direction? Why do I, and everything that exists, exist?** The way I understand it now is not that physics gives one single thing called time. Physics gives a layered structure.

In relativity, the cleanest local clock variable is proper time. The invariant spacetime interval is

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu.$$

For a timelike worldline and signature  $(-+++)$ ,

$$d\tau = \frac{1}{c} \sqrt{-ds^2}.$$

Proper time is not the  $t$ -coordinate. It is the geometric quantity accumulated by an idealized local clock carried along a path. Idealized means perfectly reliable, small enough not to disturb what is being measured, and free from practical engineering errors. In flat spacetime,

$$d\tau = dt \sqrt{1 - \frac{v^2}{c^2}}.$$

That is the core reason time dilation exists: the clock reading depends on the path through spacetime, not just on the endpoints in one coordinate frame.

A free relativistic particle follows a path that extremizes proper time. With standard conventions,

$$S = -mc \int \sqrt{-g_{\mu\nu} dx^\mu dx^\nu} = -mc^2 \int d\tau.$$

The Euler-Lagrange equations give the geodesic equation,

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} = 0.$$

So in relativity, time evolution for a free body is not motion through a fixed background time. It is worldline geometry determined by the metric connection.

## Why Time Has a Direction

The deepest puzzle is not that equations contain a time variable. The deeper puzzle is why we remember the past and not the future, why a drop of violet dye spreading through water does not gather itself back into a perfect bead, and why records accumulate in one temporal direction.

In statistical mechanics, entropy is tied to the volume of phase space compatible with a macrostate:

$$S_B = k_B \log W.$$

For a low entropy macrostate, overwhelmingly many compatible microstates evolve toward larger macroregions because those occupy vastly more volume. But this is not enough by itself. If microscopic laws are largely reversible, why do we almost always see entropy increase toward what we call the future rather than toward the past? A standard explanatory strategy is to combine statistical mechanics with a special low entropy boundary condition for the early universe, often called the **Past Hypothesis**.

For cosmology, the puzzle sharpens because the early universe was hot, smooth, and nearly homogeneous. Matter and radiation may have been close to local thermal equilibrium, but gravitational entropy was extremely low because gravity favors clumping, black hole formation, and large Weyl curvature. A smooth early Friedmann-Lemaitre-Robertson-Walker universe is therefore thermodynamically special even if the plasma was hot.

Decoherence explains a different part of the story. If a system  $S$  interacts with an environment  $E$ , the reduced state is

$$\rho_S = \text{Tr}_E(\rho_{SE}).$$

Environmental entanglement suppresses interference terms in preferred pointer bases, stabilizing records and giving effective classical branches. But decoherence alone is not a complete arrow-of-time theory, because the formation of records still depends on a low entropy boundary condition.

Most microscopic laws used in mechanics, field theory, and unitary quantum mechanics are nearly time reversal symmetric in their core form. Strictly, the Standard Model is not perfectly time reversal symmetric: charge-parity violation in the weak interaction implies time-reversal violation if charge-parity-time symmetry holds. That microscopic violation is not what explains the thermodynamic arrow, which is normally treated through statistical mechanics, records, and special boundary conditions.

## Relativity, Geometry, and Cosmic Time

The geometric core of general relativity is the metric  $g_{\mu\nu}$ , and the dynamical core is the Einstein field equation:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.$$

In matter, geometry and stress energy lock together. This is the precise mathematical form behind the idea that matter tells spacetime how to curve, and curvature tells matter how to move.

Free massive particles obey the geodesic equation, while light follows null geodesics with

$$ds^2 = 0.$$

This makes proper time local and path dependent, while coordinate time remains a matter of foliation choice. In generic curved spacetimes, there is no preferred global time function with universal physical status.

In the Arnowitt-Deser-Misner 3 + 1 formalism, spacetime is foliated into spatial slices:

$$ds^2 = -N^2 c^2 dt^2 + h_{ij}(dx^i + N^i dt)(dx^j + N^j dt),$$

where  $h_{ij}$  is the spatial metric,  $N$  is the lapse, and  $N^i$  is the shift. The crucial conceptual point is that  $t$  here is a foliation label, while lapse and shift encode gauge freedom in how slices are stacked.

Cosmology gives back a cleaner time only after imposing symmetry. In Friedmann-Lemaitre-Robertson-Walker spacetime,

$$ds^2 = -c^2 dt^2 + a(t)^2 \left( \frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right).$$

Here  $t$  is cosmic time, the proper time of comoving observers. For a comoving observer,  $dr = d\theta = d\phi = 0$ , so  $ds^2 = -c^2 dt^2 = -c^2 d\tau^2$  and  $d\tau = dt$ . Cosmic time is therefore not Newton's absolute time; it is the proper time of a special family of observers selected by large-scale homogeneity and isotropy.

The Friedmann equation is

$$H^2 = \left( \frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}.$$

It relates expansion, density, curvature, and the cosmological constant.

Horizons sharpen the distinction between local time, global time, and observable time. In Friedmann-Lemaitre-Robertson-Walker models, the particle horizon and event horizon are

$$d_p(t) = a(t) \int_0^t \frac{c dt'}{a(t')},$$

$$d_e(t) = a(t) \int_t^\infty \frac{c dt'}{a(t')}.$$

The particle horizon is about what can have influenced us by time  $t$ . The event horizon is about what we can ever influence or observe in the future. The Hubble radius  $c/H$  is important, but it is not generally identical to either horizon.

## Quantum Time and Timeless Equations

In ordinary nonrelativistic quantum mechanics, time enters as an external parameter:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle.$$

The basic architecture is a one-parameter unitary group generated by the Hamiltonian. Position, momentum, and spin are operators. Time is usually the parameter with respect to which they evolve.

The Pauli style objection matters here because it reinforces the layered view of time. If a self-adjoint operator  $\hat{T}$  were globally canonical conjugate to a semibounded Hamiltonian  $\hat{H}$ , then the unitary transformations generated by  $\hat{T}$  would shift the spectrum of  $\hat{H}$  through arbitrary real values, contradicting the lower bound:

$$[\hat{T}, \hat{H}] = i\hbar.$$

That blocks a universal canonical time operator under those assumptions. It does not block operational time observables, arrival-time positive operator-valued measures, or special restricted constructions.

The path integral formulation reframes evolution as a sum over histories:

$$K(b, a) = \int \mathcal{D}x e^{iS[x]/\hbar}.$$

This does not remove time from ordinary quantum mechanics. It rewrites time evolution in terms of amplitudes assigned to paths.

A more radical move is relational time. In idealized Page-Wootters-type constructions, a total stationary state may satisfy

$$\hat{H}_{\text{tot}} |\Psi\rangle = 0, \quad |\psi(t)\rangle \propto \langle t | \Psi \rangle,$$

while conditional subsystem states recover effective time evolution through correlations with a clock subsystem. Time becomes correlation, not background. This is elegant, but idealized. It depends on a good clock sector, appropriate factorization, and coherence.

Quantum field theory adds another layer. Relativistic causality is protected by microcausality: field operators at spacelike separation commute or anticommute, preventing controllable faster-than-light influence. A schematic scalar-field version is

$$[\phi(x), \phi(y)] = 0 \quad \text{for spacelike separation.}$$

The Feynman propagator can be written schematically as

$$D_F(x - y) = \langle 0 | T \{ \phi(x) \phi(y) \} | 0 \rangle.$$

This expression is not a simple story of one particle traveling through an ordinary background time. It is a covariant object encoding amplitudes, causal structure, and boundary prescription.

In canonical quantum gravity, the Hamiltonian constraint becomes an operator equation. The Wheeler-DeWitt equation is best treated here as schematic because the actual canonical theory contains local Hamiltonian constraints, momentum constraints, factor-ordering choices, regularization issues, and an unresolved physical inner product. In schematic form it is often written as

$$\hat{H}\Psi = 0.$$

There is no external  $t$  here. The universal state is described by a constraint, not by a time-dependent Schrodinger equation.

To recover ordinary time, one can use an approximate semiclassical form such as

$$\Psi[h, \phi] \approx A[h] e^{iMS_0[h]/\hbar} \chi[h, \phi],$$

with an approximate matter evolution equation

$$i\hbar \frac{\partial \chi}{\partial t_{\text{WKB}}} \approx \hat{H}_m \chi.$$

Here  $t_{\text{WKB}}$  is an emergent semiclassical time read off from the gravitational background, not a fundamental external variable.

## Emergent Dynamics from Static Totality

If the deepest total state is stationary, change cannot mean that the totality moves through an outside clock. The relevant question is how ordered internal change can be represented inside a globally constrained or timeless structure.

The answer I take seriously is relational. A clock is a physical subsystem whose states are correlated with another subsystem. Under the conditions of a relational model, correlations between a clock sector and another subsystem can describe ordered change without requiring an external time parameter for the whole.

A schematic form of the idea is

$$\hat{H}_{\text{tot}} |\Psi\rangle = 0,$$

while conditional states relative to a clock reading  $t$  behave as

$$|\psi(t)\rangle \propto \langle t | \Psi \rangle.$$

The total state can be stationary while internal subsystems display relative evolution. In that description, change is encoded in correlations among subsystems rather than in motion of the totality through an external clock.

Entropy gradients and stable records allow histories to acquire an ordered direction and a physical memory of earlier states. That physical description does not by itself explain the subjective experience of temporal passage; it specifies the record structure that any bridge to experience would have to respect.

On this view, the totality does not require an external clock for local time to be real. Relational time is recovered inside the total structure through correlations, records, and semiclassical or thermodynamic organization.

## Infinity and the Limits of Description

Infinity is not one thing. Mathematical infinity, ultraviolet divergence, vacuum energy divergence, singularity, and cosmic or global infinity have different meanings.

Fields assign values to spacetime points. Hilbert spaces may be infinite dimensional. Configuration spaces can be continuous. These infinities often belong to the representational language of the theory, not directly to observables.

Ultraviolet divergences appear in perturbative quantum field theory loop integrals. A representative regulated integral has the form

$$I(\epsilon) = \frac{A}{\epsilon} + B + O(\epsilon).$$

The pole in  $\epsilon$  is not interpreted as a literal measurable infinity. It is absorbed into bare parameters, after which renormalized predictions are finite and scale dependent. The renormalization group tracks this scale dependence:

$$\mu \frac{dg}{d\mu} = \beta(g).$$

Vacuum energy creates a deeper problem because gravity couples to absolute stress energy. In free field language, vacuum energy includes a formally divergent zero-point contribution:

$$\rho_{\text{vac}} \sim \frac{1}{2} \int \frac{d^3 k}{(2\pi)^3} \hbar \omega_k.$$

In nongravitational quantum field theory, only energy differences are usually observable. With gravity, the absolute scale matters. That is why the cosmological constant problem remains severe.

Singularities are different again. In general relativity, the cleaner criterion is often geodesic incompleteness, not simply a point of infinite density. A singularity theorem tells us where the classical spacetime description ceases to be extendable, not what a complete quantum gravitational theory would say.

The Schwarzschild horizon is the radius

$$r_s = \frac{2GM}{c^2}.$$

That surface is not a curvature blowup. By contrast, the central singularity is different because the Kretschmann scalar diverges there:

$$K = \frac{48G^2 M^2}{c^4 r^6}.$$

The Schwarzschild-coordinate singularity at the horizon is a coordinate artifact; the event horizon itself is a real causal boundary, not a curvature blowup. The center is an incompleteness signal for classical general relativity.

Cosmic infinity is subtler than it sounds. Even if spatial slices are globally unbounded, our observable domain is finite because the universe has finite age and finite causal horizons. Infinite universe, infinite observable universe, and every possible configuration are not the same claim.

Eternal inflation pushes infinity to its cosmological extreme. Inflating regions may continue producing new thermalized regions, potentially without bound. This creates a route to multiverse pictures and infinite ensembles of pocket universes. But it also creates the measure problem: probabilities become ambiguous when comparing infinite numbers of events, regions, or observers.

| Infinity Type          | How Physics Handles It                                  | My Reading                                         |
|------------------------|---------------------------------------------------------|----------------------------------------------------|
| Ultraviolet divergence | Regularization, renormalization, effective field theory | Usually not literal. Often a scale warning.        |
| Vacuum energy          | Renormalization plus gravity reopens the problem        | A real unresolved mismatch.                        |
| Singularity            | Geodesic incompleteness or curvature blowup             | Likely a boundary of classical general relativity. |
| Cosmic infinity        | Topology, horizons, measures, and observation limits    | Serious but not directly observable as a whole.    |

## Personal Ontology

The **Everettian interpretation** is the interpretation of quantum mechanics I find most coherent because it retains unitary evolution of the universal state and does not introduce an additional physical collapse process. Decoherence allows approximately autonomous branch structures with stable histories and records. My interpretive commitment is that those structures are real rather than merely descriptions of a single selected outcome.

On that reading, branches are distinct structures within one universal physical state. Their effective autonomy allows different local histories without making them absolutely separate realities. This structural belonging does not imply shared experience across branches.

This is not the same as saying that every logically describable form exists physically. The ontology is mode-indexed: physical branches belong to physical reality only when supported by the actual quantum state and its dynamics, while possible, impossible, mathematical, conceptual, counterfactual, and counterpossible forms belong to **Ultimate Totality** under their own modes. Physical reality is the law-compatible physical subset.

In this framework, **unified duality** preserves the distinction between the histories and records of each branch and their belonging to the universal state. Observers, where they exist, are part of those histories. Branch realism is my interpretive choice; ultimate experiential unity is an additional metaphysical commitment and does not follow from the quantum structure itself.

## Finite Hilbert Space and Emergent Locality

A physics-facing framework that sharpens this view starts from a **Hilbert space**, a quantum state, and a **Hamiltonian**. In that framing, quantization is not something nature does to itself; it is a human procedure for translating classical variables into quantum language. Nature would simply be quantum from the start, while classical physics would be the large-scale approximation that finite observers learn first.

Black-hole entropy and holographic reasoning suggest that a gravitating finite region with finite boundary area and bounded energy may have finite information capacity. If that is correct, then the infinite-dimensional

continuum used in quantum field theory is not the final ontology of nature. It is an extraordinarily successful approximation, useful inside its domain, but not necessarily the deepest description of a finite physical region.

In that kind of picture, locality is not assumed at the start. The basic ingredients are the state, the Hamiltonian, and the operator structure. The operators that appear local to us would have to emerge from how the system factors into stable subsystems. In some reconstruction programs, patterns of entanglement or mutual information can help define effective nearness, but this is model-dependent rather than a universal rule for arbitrary quantum states. In those models, spatial organization is reconstructed from relations among subsystems instead of being fully imposed from the start.

In some research programs, gravitational geometry may be recoverable from quantum-information structure under special assumptions. In approaches that connect entanglement, modular Hamiltonians, and spacetime geometry, Einstein-like gravitational behavior is not inserted as the first fact; it is recovered from deeper quantum structure under special limits. I treat that as a serious research direction, not as a completed theory.

Not every mathematical factorization of a Hilbert space defines a physical branch. Branch structure depends on dynamics, decoherence, and the stability of records; it does not require conscious observers. Environmental redundancy can make those records robust. For a history to contain an observer with memory, it additionally requires organized systems that preserve information over time. Physical organization and the presence of experience are separate questions.

## Observer Boundary and Stable Records

The partial trace can be written for many mathematical choices of subsystem, so the symbol  $X$  is not enough by itself. In my framework, a finite observer is not any random slice of Hilbert space. It is a localized, dynamically stable, record-bearing subsystem whose internal states can maintain robust, time-ordered information about its surroundings.

$$\rho_X = \text{Tr}_{\neg X}(\rho_T).$$

Here  $\rho_T$  is the state of the larger total system,  $X$  is the candidate finite subsystem, and  $\neg X$  represents everything outside that candidate subsystem. The reduced state  $\rho_X$  is what remains when the rest is traced over.

The observer boundary depends on the dynamics that sustain integrated records, memory, and, for a self-aware perspective, a self-model. The organization must resist ordinary noise through redundancy, repair, or error correction in the broad physical sense. A different mathematical partition is relevant only if it also identifies that stable organization. The boundary is functional and emergent, so it may be approximate rather than uniquely sharp at the microscopic level.

This does not solve consciousness. It only states what the physical side of a bridge theory must connect to: not a random subsystem, but a stable, embodied, record-carrying organization capable of memory, attention, and branch-local history.

## Generative Variation and Many Origins

One pattern I see everywhere is **variation**: many species, many bodies, many stars, many minds, many possible lives, and many branch-relative histories. Reality does not appear as one form repeated without difference. It appears through families, alternatives, mutations, branches, and scales.

Physics gives disciplined ways to think about plurality without pretending the whole is already mapped. Everettian quantum mechanics gives branch structure. Eternal-inflation models discuss bubble regions. Landscape reasoning discusses many possible vacuum states. Quantum-cosmology proposals discuss different boundary conditions and origin routes. These are not all proven as literal descriptions of reality, but they make a single privileged outcome look less compulsory.

When I say **ultimate totality**, I mean the whole of reality in the broadest sense: physical, possible, mathematical, conceptual, counterfactual, and counterpossible modes of form. It should not be imagined as one object sitting inside a larger outside space. It is the whole frame, so there is no external laboratory from which it can be measured as a separate object. The physically real domain is only one mode within that broader totality.

A totality without an absolute beginning or final boundary would not have to be static or uniform. It could contain regions with different ages, horizon sizes, effective laws, vacuum states, histories, and physical organizations. I do not claim every possible or impossible configuration exists physically; I claim they are contained by mode in Ultimate Totality. This does not evade past-incompleteness results for many inflationary spacetime models; it is a metaphysical possibility unless tied to a specific quantum-cosmological model.

## Bridge Schema and Classical Records

A bridge proposal must distinguish microscopic superposition from the organized physical structure that can support a personal history: decohered macroscopic records, stable pointer states, environmental redundancy, memory, embodiment, and causal order. The transition is not a physical collapse from quantum to non-quantum; it is the emergence of stable record structure for the relevant local observables.

When a local system becomes entangled with a large environment, phase relations between alternatives become practically inaccessible to the local observer. The reduced state becomes approximately diagonal in a stable record basis:

$$\rho_X \approx \sum_{\alpha} p_{\alpha} |\alpha\rangle \langle \alpha|.$$

In this notation, the  $|\alpha\rangle$  states are the record alternatives available to the finite subsystem, and  $p_{\alpha}$  are the corresponding weights. The environment does not magically destroy the universal wave function. It spreads phase information into many degrees of freedom, making interference between macroscopic alternatives effectively unavailable from inside the branch.

In this notation, the record alternatives are sufficiently stable for an approximately classical description. A system's memory within a branch preserves an ordered history of records, and the environment can store redundant information about them. This is a physical mechanism for record stability, not a derivation of subjective perspective. The connection between that structure and interiority still requires an explanation that the present framework does not provide.

This is why a lived world feels definite even if the total state is larger. The definiteness is local, relational, and record-based. It is not a proof that other branches are unreal.

## Consciousness and Agency

Decoherence and the physical description of records belong to established quantum theory. Finite local dimension, emergent locality, and recovery of time without an external parameter belong to research programs under specific assumptions. None of these results establishes my thesis of fundamental interiority or turns **Ultimate Totality** into a physically observable object.

The hard question for my view is the **bridge problem**: how does primitive interiority become organized in finite self-systems with memory, attention, embodiment, language, values, and a stable sense of being one person? Naming interiority as fundamental does not solve that. It only rejects the claim that experience emerges from a reality with no intrinsic experiential aspect anywhere.

A serious bridge theory would need constraints. It would have to respect known physics, ordinary neuroscience, decoherence, thermodynamics, and the fact that human minds depend on physical records and causal organization. If it changed quantum statistics, measurement theory, or branch weights, it would need a testable law. Without that, it is an ontology of interiority, not a new physical mechanism.

The bridge also has to be branch-local. In an Everettian universe, there is not one ordinary human ego floating above all outcomes and choosing which branches become real. There are finite self-systems inside decohered histories, each with its own records and future. A bridge theory would need to explain how a local perspective arises inside a branch without erasing the reality of counterpart perspectives in other branches.

## Objections I Take Seriously

The first objection is against **Everettian realism**. If outcomes with support in the quantum state occur in different branches, probability still requires justification from the perspective of an observer. Branch weights, self-location, and rational expectation are relevant, but naming them does not by itself solve that justification. Probability cannot be reduced to counting branches, whose boundaries are effective and approximate.

The second objection is about **ontology**. A critic can ask whether branches are truly worlds or only useful descriptions after decoherence. I take that seriously. My answer is that if the universal wave function is real and no physical collapse removes the other terms, treating the other decohered branches as unreal feels less coherent than accepting them as real parts of the total state.

The third objection is against **unified duality**. Physical relations among systems do not prove experiential unity. My claim of global belonging also does not identify each local system with the totality: systems keep their own records, boundaries, and histories. The thesis of an ultimate shared interiority requires an additional metaphysical argument.

The fourth objection is about the interiority commitment. Someone can say that atoms, molecules, and inanimate matter behave the way they do because of physical law, not because consciousness is present in them. I accept the public description. My claim is not a new force and not a replacement for physics. It is a claim about the intrinsic nature of lawful reality, and the bridge theory remains unfinished.

The fifth objection is **free will**. A unitary Everettian universe does not give me libertarian freedom in the sense of an uncaused chooser standing outside physics. If I use the phrase free will, I mean branch-level agency: deliberation, attention, memory, values, inhibition, and self-control are real causal processes inside a finite self-system.

Branch-level agency means the agent does not choose which branches exist globally. The agent is the branch-local process by which reasons, memories, emotions, and values become correlated with later records and actions. Nearby counterpart agents may make different choices in other branches, but that does not make deliberation fake inside this branch. It rejects outside-the-wavefunction choice while preserving agency as a real pattern of reasoning and self-control inside a causal history.

## Personal Synthesis

My criterion for distinguishing infinities centers on the stability of observables. An infinity is probably a **representational artifact** when changing the regulator, cutoff, coordinates, or idealization leaves finite observable predictions stable within the theory's domain. It is an **incompleteness signal** when those predictions cannot be stabilized without new physics, new boundary data, or replacement of the relevant invariant structure. This criterion does not by itself decide whether global physical infinities exist.

More formally: an infinity is an artifact when it can be removed or absorbed into scale-dependent parameters without changing observable dimensionless ratios  $R_i(\mu)$  at the scale  $\mu$  where the theory is being tested. It is an incompleteness signal when predictions at  $\mu$  cannot be made stable without new physics, new boundary data, or a replacement of the theory's invariant structure.

In renormalizable quantum field theories, many ultraviolet divergences can be absorbed into scale-dependent parameters while retaining control of finite observables. **Gravitational singularities** pose a different problem: they can signal that the classical spacetime description is incomplete. The cosmological constant problem is also not merely the removal of a divergence; it includes the mismatch between quantum-vacuum estimates and the observed gravitational effect.

For experienced time, I distinguish the physical conditions of reports, memory, anticipation, and ordering judgments from the subjective presence of a sequence. Geometry, entropy gradients, and integrated records can describe those functions, but adding the word interiority does not explain their connection to experience. If two systems physically identical under the current description could differ in lived temporal flow, one would have to specify what psychophysical principle permits that difference. If it had observable consequences, it would also require a testable physical account. I do not know whether such a difference is possible; the bridge problem remains open.

## Personal Vision Appendix

This appendix belongs to the level of my personal metaphysical position. I believe that **Ultimate Totality** contains finite perspectives and, through them, knows limitation, contrast, memory, pain, beauty, and variation. I use **learn** as a characterization of that belief, not as a measurable biological or computational process of the totality.

The scientifically disciplined version is weaker: finite observers produce records, compare states, store memories, and occupy branch-relative histories. My personal vision adds that these finite perspectives are not meaningless fragments. They are how a deeper reality becomes concrete from within without being exhausted by any one life, world, or branch.

## Axioms of Perspective

These are not new physical laws. They are the clearest way I can state the philosophical rules that organize this framework without leaving them scattered through the prose.

| Axiom         | Statement                                                                                                                                  | Boundary                                                                                   |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Totality      | Ultimate Totality contains every mode of form; the physical subset is the law-compatible physical domain.                                  | This is my metaphysical framework, not a laboratory measurement of the whole from outside. |
| Perspective   | Local physics is described relative to a finite subsystem capable of carrying stable records.                                              | A random mathematical factorization is not enough to define an observer.                   |
| Records       | A finite observer must maintain robust, time-ordered records through memory, embodiment, causal organization, and environmental stability. | This is a functional boundary, not a sharp wall drawn in empty space.                      |
| Bridge        | Subjective linear history maps onto the persistent branch-local record structure available to that finite perspective.                     | This does not yet solve consciousness; it constrains what a bridge theory must connect to. |
| Relative Time | Time is recovered through correlations between subsystems, especially clocks, records, entropy gradients, and semiclassical geometry.      | The totality does not need an outside clock for local time to be real inside it.           |

## Open Questions

- Why was the early universe in such a low entropy condition?
- Was the early universe special because of low gravitational entropy, low Weyl curvature, or a deeper boundary principle?
- Is the Past Hypothesis a law, a boundary condition, or an explanatory postulate?
- How should gravitational entropy be defined generally?
- Is time fundamental or emergent?
- What is the correct quantum gravity framework?
- Can relational time survive realistic imperfect clocks?
- How exactly should the boundary of a finite observer be defined through stable records?
- What threshold turns microscopic entanglement into branch-local classical records?
- Are Everettian branches fully real worlds or emergent structures inside one universal state?
- Is variation only a biological and cosmological pattern, or a deeper generative feature of existence itself?
- Which infinities are artifacts, and which infinities signal incompleteness?
- Are global infinities physically meaningful or observationally inaccessible?
- Can finite self-systems ever experience the totality without erasing local difference?
- Can physics explain consciousness itself, or will it need a theory of intrinsic experience in addition to the structures correlated with it?
- Can consciousness, meaning, death, and first-person presence ever be exhausted by mathematics, physics, computation, or laboratory testing?

## Final Synthesis

Proper time and thermodynamic records have an established physical role; relational and semiclassical recovery of time are studied under specific assumptions. This makes it possible to consider local time real without presupposing a fundamental external parameter. Infinities must be separated into representational artifacts, incompleteness signals, and possible global physical properties rather than treated as one category.

My position combines Everettian realism, **unified duality**, fundamental interiority, and generative **variation**. Everettian realism is an interpretation of quantum mechanics; the remaining claims express my metaphysical synthesis. I defend the view that finite configurations are locally distinct while belonging to a larger structure, without claiming that this belonging solves how first-person experience exists.

A condensed formal summary is:

$$\mathcal{T} = \bigcup_{m \in \mathcal{M}} \mathcal{F}_m,$$

where  $\mathcal{T}$  is Ultimate Totality,  $\mathcal{M}$  is the set of modes, and  $\mathcal{F}_m$  is the set of forms belonging to mode  $m$ .

$$\mathcal{M} = \{\text{physical, possible, mathematical, conceptual, counterfactual, counterpossible}\}.$$

$$\mathcal{F}_{\text{physical}} \subset \mathcal{T}, \quad \mathcal{F}_{\text{physical}} = \{x \in \mathcal{T} : x \text{ is compatible with physical law}\}.$$

$$x \in \mathcal{T} \not\Rightarrow x \in \mathcal{F}_{\text{physical}}.$$

So containment in Ultimate Totality is broader than physical existence. Physical reality is the law-compatible subset; possible, mathematical, conceptual, counterfactual, and counterpossible forms belong by mode, not by physical instantiation.

## Glossary

- **Coordinate time:** A time label used inside a chosen coordinate system. It is useful, but it is not automatically what a physical clock measures.
- **Proper time:** The time measured by an idealized local clock moving along a particular path through spacetime. Idealized means perfectly reliable, small enough not to disturb what is being measured, and free from practical engineering errors.
- **Cosmic time:** The shared clock time used in highly symmetric cosmological models, measured by observers moving with the cosmic expansion.
- **Thermodynamic time:** The direction of time associated with entropy growth, stable records, memory, and irreversible macroscopic change.
- **Relational time:** Time understood through correlations between physical systems rather than as an external background clock.
- **Semiclassical time:** Time that appears when part of a quantum gravitational system behaves almost like a classical spacetime background.
- **Hilbert space:** The mathematical space where quantum states live.
- **Hamiltonian:** The operator or function that represents the energy of a system and usually generates its time evolution.
- **Decoherence:** The process where interaction with the environment suppresses visible quantum interference, making branches look classical and stable from inside.
- **Stable record:** Physical information that remains robust enough to support memory, evidence, history, or branch-local identity.
- **Observer boundary:** The functional boundary of a finite perspective, defined by stable records, causal organization, embodiment, and memory rather than by an arbitrary mathematical cut.
- **Everettian multiverse:** The interpretation of quantum mechanics where the universal wave function evolves without collapse, and different outcomes exist as real branches of one total state.
- **Branch:** A stable, effectively separated part of the quantum state after decoherence, experienced from inside as one outcome or one world history.
- **Finite configuration:** A local form inside reality, such as an animal, plant, star, planet, person, memory, branch, or world. It is real locally, but not absolutely separate from the totality.
- **Unified duality:** My label for the view that local distinction and global belonging are both real.
- **Mutual information:** A measure of how much knowing one subsystem reduces uncertainty about another.
- **Emergent locality:** The idea that spatial nearness may arise from quantum information structure rather than being a primitive feature of reality.
- **Regularization:** A mathematical method for temporarily controlling infinities so calculations can be handled clearly.
- **Renormalization:** The process of absorbing controlled infinities into measured parameters so the theory makes finite predictions.
- **Artifact infinity:** An infinity that can be removed by a better regulator, coordinate choice, or renormalization without changing stable observable predictions.
- **Incompleteness signal:** An infinity that marks a breakdown of the theory's domain and requires new physics, new boundary data, or a replacement framework.
- **Measure problem:** The problem of defining probabilities when a theory produces infinitely many regions, branches, events, or observers.
- **Ultimate totality:** The whole of reality in the broadest sense, including physical, possible, mathematical, conceptual, counterfactual, and counterpossible modes of form. I use it metaphysically, not as an established cosmological object.

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